

ISyE 6644A/MSA — Spring 2025 — Test 3 Solutions

(Revised 4/30/25)

This is a take-home test. You have until **Friday, May 2** to do it. The test is designed to take at most 3 hours, so don't spend *tooooo* much time on it. **Obviously, don't talk to anyone else about the test except for me. And no ChatGPT.** All questions are 3 points, except for the last question, which is 1 point. Good luck!

Show all work, and make sure everything is very neat and well-organized. CLEARLY indicate your answers.

1. Suppose the continuous random variable X has p.d.f. $f(x) = 5x^4$ for $0 \leq x \leq 1$. Find the inverse of X 's c.d.f., and thus show how to generate the RV X in terms of a Unif(0,1) PRN U .

Solution: The c.d.f. is easily shown to be $F(x) = x^5$ for $0 \leq x \leq 1$, so that the Inverse Transform Theorem gives $F(X) = X^5 = U \sim \text{Unif}(0, 1)$. Solving for X , we obtain the desired inverse, $F^{-1}(U) = X = U^{1/5}$. $\square$

2. Suppose the continuous random variables X_1 and X_2 are i.i.d. with p.d.f. $f(x) = 2x$ for $0 \leq x \leq 1$. Let $Y = \min(X_1, X_2)$. Find the inverse of Y 's c.d.f., and thus show how to generate the RV Y in terms of a single Unif(0,1) PRN U . (This may be a little tricky, but be patient.)

Solution: First of all,

$$\begin{aligned}
 F_Y(y) &\equiv P(Y \leq y) \\
 &= P(\min(X_1, X_2) \leq y) \\
 &= 1 - P(\min(X_1, X_2) > y) \\
 &= 1 - [P(X_1 > y)]^2 \\
 &= 1 - \left[\int_y^1 2x \, dx \right]^2 \\
 &= 1 - [x^2|_y^1]^2 \\
 &= 2y^2 - y^4.
 \end{aligned}$$

Setting $F_Y(Y) = 2Y^2 - Y^4 = U$ and solving, we obtain (eventually)

$$Y = \sqrt{1 - \sqrt{1 - U}}. \quad \square$$

Remark: It's also perfectly OK to use $Y = \sqrt{1 - \sqrt{1 - U}}$ $\square$

3. Suppose that $U_1 = 0.15$, $U_2 = 0.65$, $U_3 = 0.82$, $U_4 = 0.06$, and $U_5 = 0.26$. Use our acceptance-rejection technique from class to generate a $\text{Pois}(\lambda = 3.5)$ random variate. (You may not need to use all of the uniforms.)

Solution: Define $p_n \equiv \prod_{i=1}^{n+1} U_i$. We'll stop as soon as $p_n < e^{-3.5} = 0.0302$. Let's make the following convenient table.

n	U_{n+1}	p_n	Stop?
0	0.15	0.15	nope
1	0.65	0.0975	nope
2	0.82	0.0800	nope
3	0.06	0.0048	yup

So we take $N = 3$. $\square$

4. Consider a $\text{Pois}(300)$ random variable N . Show how to use a normal approximation with exactly one $\text{Unif}(0,1)$ PRN U generate a realization of N ; and demonstrate your technique with $U = 0.6$.

Solution: By the Central Limit Theorem, $N \approx \text{Nor}(\lambda, \lambda) \sim \text{Nor}(300, 300)$, and so

$$Z \equiv \frac{N - 300}{\sqrt{300}} = \frac{N - 300}{17.321} \approx \text{Nor}(0, 1).$$

Thus, applying a continuity correction (and using the fact a Poisson RV cannot be negative), we can take

$$N = \max \left\{ 0, \lfloor 300 + 17.321 \text{Nor}(0, 1) + 0.5 \rfloor \right\} = \max \left\{ 0, \lfloor 300 + 17.321 \Phi^{-1}(U) + 0.5 \rfloor \right\}. \quad \square$$

For $U = 0.6$, we have $\Phi^{-1}(0.6) = 0.2533$, and so

$$N = \max \left\{ 0, \lfloor 300 + 17.321(0.2533) + 0.5 \rfloor \right\} = \max \left\{ 0, \lfloor 304.89 \rfloor \right\} = 304. \quad \square$$

5. Show how to generate in Arena a discrete random variable X for which we have $P(X = -3) = 0.3$, $P(X = 3.5) = 0.6$, and $P(X = 4) = 0.1$.

Solution: `DISC(0.3, -3, 0.9, 3.5, 1.0, 4)`. $\square$

6. YES or NO? In our Arena Call Center example, is it possible that entities are left in the system when it shuts down at 7:00 p.m. (even though we stopped allowing customers to enter the system at 6:00 p.m.)?

Solution: YES — because of the small chance that a callback will occur. $\square$

7. YES or NO? In our Arena Inventory Process example, were we able to accommodate backlogs, arbitrary lead times, different customer arrival patterns, and multiple inventory checks per day?

Solution: YES. $\square$

8. TRUE or FALSE? In our Arena Manufacturing Process example, we defined a “vector” of cell visitation sequences with the help of the **Advanced Set** functionality of the **Advanced Process** panel.

Solution: TRUE. $\square$

9. TRUE or FALSE? In our Arena Manufacturing Process example, it is possible for the same entity to be scheduled to visit the *same* cell twice, with *different* service time distributions on the two visits!

Solution: TRUE. $\square$

10. It’s GIGO time! Let’s consider an $M/M/1$ queueing system with $\text{Exp}(\lambda)$ interarrivals and $\text{Exp}(\mu)$ FIFO services at a single server. You may recall from some class (either this one or stochastic processes) that the steady-state expected cycle time (i.e., the time that the customer is in the system, including wait + service) is $w = 1/(\mu - \lambda)$.

If you were to try this out in Arena, let's say with **EXP0**($10 = 1/\lambda$) interarrivals and **EXP0**($8 = 1/\mu$) services (note the notation change between my usual "Exp" and Arena's "EXP0"), then we'd get $w = 1/(0.125 - 0.1) = 40$. Go ahead, see for yourself in Arena, but make sure that you run the system for 100,000 or so customers so that you can be sure that you're in steady-state!

Finally, here's the GIGO question, which will show what can happen when you mis-model a component of your process: What is the (approximate) steady-state expected cycle time if you have i.i.d. **UNIF**(5, 15) interarrivals instead of **EXP0**($10 = 1/\lambda$) interarrivals? Note that both interarrival distributions have the same mean (10), but that doesn't necessarily imply that they'll have the same expected cycle times. Hint: You may want to use Arena as described above.

- (a) about 1
- (b) about 10
- (c) about 23
- (d) about 40
- (e) about 62

Solution: (c). The **UNIF** case has waaaay smaller tails than the **EXP0**, so it's reasonable to assume that the cycle times will tend to be lower for the **UNIF** case. In fact, after 100,000 customers in Arena, I got an average time of 23.5. Thus, (c) is the right answer. $\square$

11. If U_1 and U_2 are i.i.d. **Unif**(0,1) with $U_1 = 0.45$ and $U_2 = 0.45$, use Box-Muller to generate two i.i.d. **Nor**(0,1) realizations. (I was lazy and made U_1 and U_2 the same value so that the problem will be easier for me to grade! But note that you should still get two different Z_1 and Z_2 values.)

Solution: We have

$$\begin{aligned} Z_1 &= \sqrt{-2\ln(U_1)} \cos(2\pi U_2) = \sqrt{-2\ln(0.45)} \cos(0.9\pi) = -1.2019 \\ Z_2 &= \sqrt{-2\ln(U_1)} \sin(2\pi U_2) = \sqrt{-2\ln(0.45)} \sin(0.9\pi) = 0.3905. \quad \square \end{aligned}$$

12. Using the two normals that you came up with from Question 11, generate a Cauchy random variable.

Solution: Either $Z_1/Z_2 = -3.0778$ or $Z_2/Z_1 = -0.3249$. $\square$

13. Suppose that Z_1, Z_2, Z_3 , and Z_4 are i.i.d. $\text{Nor}(0,1)$ random variables, and let

$$T = \frac{Z_1}{\sqrt{(Z_2^2 + Z_3^2 + Z_4^2)/3}}.$$

Find the value of c such that $P(T < c) = 0.99$.

Solution: Note that

$$T \sim \frac{\text{Nor}(0,1)}{\sqrt{\chi^2(3)/3}} \sim t(3).$$

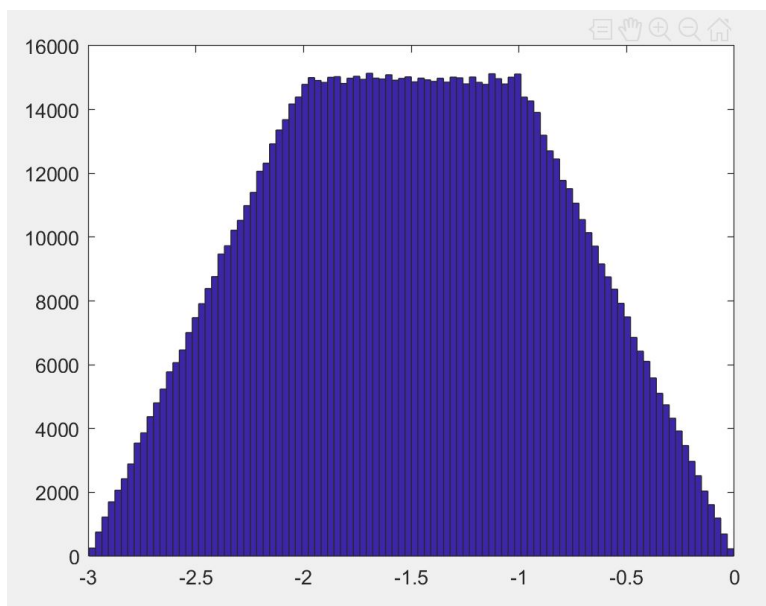
Therefore,

$$0.99 = P(T < c) = P(t(3) < c),$$

so that the quantile $c = t_{0.01,3} = 4.541$. $\square$

14. Suppose that $U \sim \text{Unif}(0,1)$ and $V \sim \text{Unif}(1,3)$ and that U and V are independent. What does the distribution of $U - V$ look like? (Don't try to do this analytically...just simulate many thousands of times.)

Solution: A trapezoid from -3 to 0 , as illustrated in the figure (that I obtained from 1,000,000 replications). $\square$



15. Suppose X has the Laplace distribution with p.d.f. $f(x) = \frac{\lambda}{2}e^{-\lambda|x|}$ for $x \in \mathbb{R}$ and $\lambda > 0$. Which of the methods below would be very reasonable to use to generate realizations from this distribution? (There may be more than one correct answer.)
- (a) Ask a UGA student.
 - (b) Inversion.
 - (c) Acceptance-Rejection.
 - (d) Composition.

Solution: (a) is a sick joke; (b) is OK because we can get the c.d.f. in an easy closed form (see the composition notes); (c) is OK because the p.d.f. is nice and finite; (d) is OK because this is the example we did in class. $\square$

16. Consider a bivariate normal random variable (X, Y) , for which $E[X] = -3$, $\text{Var}(X) = 4$, $E[Y] = 2$, $\text{Var}(Y) = 9$, and $\text{Cov}(X, Y) = 2$. Find the Cholesky matrix associated with (X, Y) , i.e., the lower-triangular matrix C such that $\Sigma = CC'$, where Σ is the variance-covariance matrix.

Solution: By class notes, we saw that for $k = 2$, we obtain

$$C = \begin{pmatrix} \sqrt{\sigma_{11}} & 0 \\ \frac{\sigma_{12}}{\sqrt{\sigma_{11}}} & \sqrt{\sigma_{22} - \frac{\sigma_{12}^2}{\sigma_{11}}} \end{pmatrix} = \begin{pmatrix} \sqrt{4} & 0 \\ \frac{2}{\sqrt{4}} & \sqrt{9 - \frac{2^2}{4}} \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 1 & 2\sqrt{2} \end{pmatrix}. \quad \square$$

17. Suppose $Z_1 = 1$ and $Z_2 = 2$ are i.i.d. $\text{Nor}(0,1)$ realizations. Using the notation from and your answer to Question 16, generate (X, Y) .

Solution: By class notes, we have

$$\begin{aligned} X &= E[X] + c_{11}Z_1 = -3 + 2(1) = -1 \\ Y &= E[Y] + c_{21}Z_1 + c_{22}Z_2 = 2 + 1(1) + 2\sqrt{2}(2) = 8.657. \quad \square \end{aligned}$$

18. Consider a nonhomogeneous Poisson arrival process with rate function $\lambda(t) = 2t$ for $t \geq 0$. Find the probability that there will be exactly 2 arrivals between times $t = 1$ and 2.

Solution: The distribution of the number of arrivals between times 1 and 2 is

$$N(2) - N(1) \sim \text{Pois} \left(\int_1^2 \lambda(t) dt \right) \sim \text{Pois} \left(\int_1^2 2t dt \right) \sim \text{Pois}(3).$$

Thus,

$$P(N(2) - N(1) = 2) = P(\text{Pois}(3) = 2) = \frac{e^{-3}3^2}{2!} = 0.2240. \quad \square$$

19. Suppose we are generating arrivals from a nonhomogeneous Poisson process with rate function $\lambda(t) = 1 + \sin(\pi t)$, so that the maximum rate is $\lambda^* = 2$, which is periodically achieved. Suppose that we generate a *potential* arrival (i.e., one at rate λ^*) at time $t = 0.85$. What is the probability that our usual thinning algorithm will actually accept that potential arrival as an actual arrival? (Note that the π means that calculations are in radians.)

Solution: The thinning algorithm says that we accept with probability

$$\frac{\lambda(t)}{\lambda^*} = \frac{1 + \sin(0.85\pi)}{2} = 0.727. \quad \square$$

20. Suppose $X_1, X_2, \dots$ is an i.i.d. sequence of random variables with mean μ and variance σ^2 . Consider the process $Y_n(t) \equiv \sum_{i=1}^{\lfloor nt \rfloor} (X_i - \mu) / (\sigma\sqrt{n})$ for $t \geq 0$. What is the asymptotic probability that $Y_n(4)$ will be at least -2 as n becomes large?

Hint: Recall that Donsker's Theorem states that $Y_n(t)$ converges to a standard Brownian motion as n becomes large.

Solution: The Hint reminds us that as $n \rightarrow \infty$, we have

$$Y_n(t) \xrightarrow{\mathcal{D}} \mathcal{W}(t) \sim \text{Nor}(0, t).$$

In particular, $Y_n(4) \xrightarrow{\mathcal{D}} \text{Nor}(0, 4)$, so that

$$P(Y_n(4) \geq -2) \approx P\left(\text{Nor}(0, 1) \geq \frac{-2-0}{\sqrt{4}}\right) = P(\text{Nor}(0, 1) \geq -1) = 0.8413. \quad \square$$

21. Which of the following properties of a Brownian motion process $\mathcal{W}(t)$ are true? More than one answer may be correct.

- (a) $\mathcal{W}(0) = 0$.
- (b) $\mathcal{W}(t) \sim \text{Nor}(0, t)$.
- (c) $\mathcal{W}(3) - \mathcal{W}(1)$ has the same distribution as $\mathcal{W}(11) - \mathcal{W}(9)$.
- (d) $\mathcal{W}(3) - \mathcal{W}(1)$ is independent of $\mathcal{W}(4) - \mathcal{W}(2)$.
- (e) $\text{Cov}(\mathcal{W}(3), \mathcal{W}(5)) = 3$.

Solution: (a) and (b) are TRUE because they are fundamental axioms of Brownian motion. (c) is TRUE by stationary increments (another fundamental axiom). (d) is FALSE because it violates independent increments (due to overlapping time intervals). (e) is TRUE since we showed in class that $\text{Cov}(\mathcal{W}(a), \mathcal{W}(b)) = \min(a, b)$.
□

22. Find the sample variance of . .

- (a) The 10 observations 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.
- (b) If you have time, the n observations $1, 2, \dots, n$.

Solution: Let's do the general case. First, it is well-known that

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \frac{1}{n} \sum_{i=1}^n i = \frac{n(n+1)}{2n} = \frac{n+1}{2}.$$

Thus,

$$\begin{aligned} S^2 &= \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n X_i^2 - n\bar{X}^2 \right) \quad (\text{from class notes}) \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n i^2 - n \left(\frac{n+1}{2} \right)^2 \right) \quad (\text{from previous work}) \\ &= \frac{1}{n-1} \left(\frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)^2}{4} \right) \quad (\text{another famous result}) \\ &= \frac{n+1}{12(n-1)} \left(2n(2n+1) - 3n(n+1) \right) \quad (\text{easy algebra}) \\ &= \frac{n(n+1)}{12}. \end{aligned}$$

In summary, the answer to (a) is $110/12 = 9.167$ and (b) is $n(n+1)/12$. $\square$

23. If $X_1, \dots, X_{10}$ are i.i.d. $\text{Exp}(1/7)$ (i.e., having mean 7), what is the expected value of the sample variance S^2 ?

Solution: S^2 is always unbiased for the variance of X_i . Thus, we have $E[S^2] = \text{Var}(X_i) = 1/\lambda^2 = 49$. $\square$

24. TRUE or FALSE? The mean squared error of an estimator is the square of the bias plus the square of its variance.

Solution: FALSE. It's $\text{bias}^2 + \text{var}$. $\square$

25. If $X_1 = 1$, $X_2 = -3$, and $X_3 = -1$ are i.i.d. realizations from a $\text{Nor}(\mu, \sigma^2)$ distribution, what is the value of the maximum likelihood estimate for the variance σ^2 ?

Solution: We know from a class example that

$$\hat{\sigma}^2 = \frac{n-1}{n} S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 = 2.667. \quad \square$$

26. Suppose that we take three i.i.d. observations $X_1 = 3$, $X_2 = 4$, and $X_3 = 3$ from $X \sim \text{Exp}(\lambda)$. Using the maximum likelihood estimate for λ , find the MLE of $P(X > 2)$.

Solution: By class notes, the MLE of λ is $\hat{\lambda} = 1/\bar{X} = 3/10$. By the Invariance Property of MLEs, we have

$$\hat{P}(X > 2) = e^{-\hat{\lambda}x} = e^{-(3/10)(2)} = e^{-0.6} = 0.549. \quad \square$$

27. Suppose we're conducting a χ^2 goodness-of-fit test to determine whether or not 100 i.i.d. observations are from a *Johnson* distribution with $s = 4$ unknown parameters a , b , c , and d . (The Johnson distribution is very general and often fits data quite well.) If we divide the observations into $k = 10$ equal-probability intervals and we observe a g-o-f statistic of $\chi_0^2 = 14.2$, will we ACCEPT (i.e., fail to reject) or

REJECT the null hypothesis of the Johnson? Use level of significance $\alpha = 0.05$ for your test.

Solution: Note that the χ^2 test has $\nu = k - s - 1 = 10 - 4 - 1 = 5$ degrees of freedom. Then

$$\chi_0^2 = 14.2 > \chi_{0.05,5}^2 = 11.07,$$

so we (easily) REJECT. $\square$

28. Let's run a simulation whose output is a sequence of consecutive customer waiting times in a crowded store. Which of the following statements are true (there may be more than one correct answer)?

- (a) The waiting times are independent.
- (b) The waiting times are correlated.
- (c) The waiting times are normally distributed.
- (d) The waiting times may not be identically distributed.

Solution: Just (b) and (d). Consecutive waiting times are rarely i.i.d. normal. $\square$

29. Suppose we want to estimate the expected average waiting time (in minutes) for the first $m = 100$ customers at a bank. We make $r = 3$ independent replications of the system, each initialized empty and idle and consisting of 100 waiting times. The resulting replicate means are:

i	1	2	3
$\bar{Z}_i$	2	4	1

Find a 95% two-sided confidence interval for the mean average waiting time for the first 100 customers.

Solution: The sample mean and sample variance of the 3 replicate means are easily calculated as $\bar{\bar{Z}}_3 = 2.333$ and $S_{\bar{Z}}^2 = 2.333$ (amazingly, they're the same!). For

level $\alpha = 0.05$, we have $t_{\alpha/2, r-1} = t_{0.025, 2} = 4.303$, and so the CI is:

$$\begin{aligned}\mu &\in \bar{Z}_r \pm t_{\alpha/2, r-1} \sqrt{\frac{S_Z^2}{r}} \\ &= 2.333 \pm 4.303 \sqrt{\frac{2.333}{3}} \\ &= 2.333 \pm 3.795 \\ &= [-1.462, 6.128]. \quad \square\end{aligned}$$

Actually, since we're talking about times here, it would've been OK to use the answer $\mu \in [0, 6.128]$. $\square$

30. Suppose that $\mu \in [-30, 90]$ is a 95% confidence interval for the mean cost incurred by a certain inventory policy. Further suppose that this interval was based on 4 independent replications of the underlying inventory system. Unfortunately, the boss has decided that she wants a 90% confidence interval. Can you supply it?

Solution: The 95% confidence interval is of the form

$$[-30, 90] = \bar{X} \pm t_{\alpha/2, b-1} y = 30 \pm t_{0.025, 3} y,$$

where the half-length $t_{0.025, 3} y = 60$.

The new 90% CI will therefore be of the form

$$\begin{aligned}\bar{X} \pm t_{0.05, 3} y &= 30 \pm \frac{t_{0.05, 3}}{t_{0.025, 3}} t_{0.025, 3} y \\ &= 30 \pm \frac{2.353}{3.182} 60 \\ &= 30 \pm 44.37 \\ &= [-14.37, 74.37]. \quad \square\end{aligned}$$

31. TRUE or FALSE? Welch's method is a graphical technique to estimate truncation (initialization bias) points for steady-state simulation.

Solution: TRUE. $\square$

32. Suppose that we're studying a stochastic process whose covariance function is $R_i = 3 - |i|$ for $i = 0, \pm 1, \pm 2$, and 0 otherwise. Find the variance of $\bar{X}_5$ (the sample mean of the first 5 observations).

Solution: From class notes, we have the following famous equation for the variance of the sample mean.

$$\begin{aligned}
 \text{Var}(\bar{X}_n) &= \frac{1}{n} \left[R_0 + 2 \sum_{i=1}^{n-1} \left(1 - \frac{i}{n} \right) R_i \right] \\
 &= \frac{1}{5} \left[3 + 2 \sum_{i=1}^4 \left(1 - \frac{i}{5} \right) R_i \right] \\
 &= \frac{1}{5} \left[3 + 2 \sum_{i=1}^2 \left(1 - \frac{i}{5} \right) R_i \right] \quad (\text{since } R_3 = R_4 = 0) \\
 &= \frac{1}{5} \left[3 + 2 \left(\frac{4}{5} \right) 2 + 2 \left(\frac{3}{5} \right) 1 \right] \\
 &= 1.48 \quad \square
 \end{aligned}$$

33. Consider a particular data set of 100000 stationary waiting times obtained from a large queueing system. Suppose your goal is to get a confidence interval for the unknown mean. Would you rather use (a) 50 batches of 2000 observations or (b) 10000 batches of 10 observations each?

Solution: Recall that the method of batch means requires a very large batch size; and, further, the notes sort of recommend at least 30 batches, (but not necessarily more at the cost of smaller-than-needed batch size). Thus, (a) is the way to go! $\square$

34. You're stranded on a desert island and can only listen to one thing on YouTube. What's it gonna be?

- (a) The Zombies singing their Rock and Roll Hall of Fame hits.
- (b) Justin Bieber
- (c) REO Speedwagon
- (d) REO Bieberwagon

Solution: You'd better say (a). $\square$